

Scrapping and EDA for SkinnyTaste

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1. INTRODUCTION

The focus of this mini-project was to scrape recipes from the website www.skinnytaste.com for the purpose of learning scraping and exploratory data analysis (EDA). As part of this project, I aimed to scrape 50 pages, each containing several recipes, and extract relevant information from the site for further analysis. This task primarily involved collecting data from various recipe pages and storing key features that could be used for EDA. If time allows in or after this course, future steps may include applying machine learning models to the dataset, such as clustering or classification, to derive deeper insights about the recipes or user preferences.

2. Data Description

For the data collection process, I focused on obtaining six key features for each recipe to ensure a comprehensive analysis. The first feature, the name of the food, is stored as a string and serves as the unique identifier for each recipe. The second feature, the image of the food, is captured as a string containing the URL of the recipe's image, providing a visual representation that enhances the user interaction component. The third feature, calories, is a floating-point value that gives a precise numerical representation of the energy content in each dish, which is essential for dietary analysis. The fourth feature, personal points, is an integer value representing the points assigned to the recipe based on various nutritional factors (This is my assumption). The fifth feature, the summary, is a string that provides a concise description of the recipe, offering an overview of its preparation, ingredients, or other noteworthy aspects. Lastly, the recipe key is a string in which keys are joined by “,” that classifies the recipe into various

dietary and nutritional categories, such as “GF” (Gluten-Free), “LC” (Low-Carb), “Q” (Quick), and “V” (Vegetarian). These categories offer additional insights into the dietary preferences and restrictions that the recipe caters to. By capturing these six features, the dataset provides valuable information for both exploratory data analysis (EDA) and future machine learning tasks. Each feature contributes uniquely to the overall understanding of the dataset, enabling a deeper investigation into nutritional patterns and user preferences.

3. Proposed Approach

The approach to scraping the website www.skinnytaste.com was structured and carefully designed to minimize complexity while ensuring the collection of relevant data. I opted to use the BeautifulSoup library in Python for this task due to its simplicity and ease of use, particularly given that the website did not require any JavaScript interaction. BeautifulSoup allowed me to parse HTML pages effectively without the overhead of a headless browser, making it a lightweight and efficient choice for this project. Alternative tools like Selenium and Scrapy were considered, but I chose to avoid Selenium due to the lack of need for dynamic JavaScript handling, and Scrapy was skipped due to its higher complexity for a relatively straightforward scraping task.

The scraping process itself was divided into two main stages. First, I collected all the URLs for the individual recipe pages by navigating through the pagination system on the website. This approach enabled me to gather a list of 938 unique recipe URLs, which were

stored in a CSV file for further use. By saving the URLs first, I was able to break the scraping task into manageable steps, ensuring that I could easily revisit the data collection process if any issues arose later. In the second stage, I scraped each recipe individually, extracting the six target features described earlier. This method allowed me to focus on gathering clean and structured data for each recipe while ensuring I captured only the most relevant information, such as recipes with available calorie counts, personal points, and recipe keys. This stepwise approach significantly improved the efficiency of the scraping process, allowing me to build a comprehensive dataset.

4. Challenges Faced

Several challenges were encountered during the data collection process, most of which were related to distinguishing between relevant recipes and non-recipe content such as articles. The first major issue arose when dealing with the pagination. The website contains both recipe and article posts, which made it difficult to directly target only the recipe pages. Initially, I attempted to filter out articles by looking for the presence of "personal points" and "Recipe Key" attributes, as articles generally did not contain these features. However, this method was not entirely reliable, as some recipes were ignored in this filtration. I then experimented with only the "personal points" features as filters, and this approach worked. To ensure more accuracy, I eventually collected URLs for both recipes and articles and performed a manual cross-verification step.

Another challenge involved scraping the recipe summaries, as there was no clearly defined class or identifier for the summary on each recipe page. This required me to experiment with targeting the parent elements around the summary content, and after some trial and error, I was able to successfully extract this information. Additionally, scraping the images for the recipes proved challenging due to the lack of a unique selector for the image URLs. To resolve this, I extracted the image URL directly from the recipe index page where the recipe URLs were initially gathered, which provided a more straightforward solution to this problem. These challenges underscored the importance of flexibility and adaptability when dealing with unstructured web data and highlighted the need for

thoughtful, step-by-step refinement of the scraping process.

5. Data Cleaning

Once the data was collected, I undertook a data cleaning process to ensure the dataset was suitable for exploratory data analysis. One of the key issues I faced was missing values in the recipe key feature, which is essential for the distribution analysis of recipe keys. Recipes that lacked a recipe key were dropped from the dataset to avoid skewing the analysis. Since the recipe key was a string so first I converted it into a list and then, I used the `.explode()` function to separate these values for more granular analysis during EDA. After cleaning the data, I was left with 928 valid recipes, which provided a solid foundation for further analysis. This cleaning process was essential for maintaining the integrity of the dataset and ensuring accurate results in the subsequent EDA phase.

6. Exploratory Data Analysis (EDA)

With the cleaned dataset in hand, I proceeded with exploratory data analysis (EDA) to uncover patterns and insights from the recipe data. The first step was analyzing the calories distribution across the dataset, which provided a clear picture of the range of energy content in the recipes. This analysis revealed clusters of low-calorie and high-calorie recipes, highlighting potential trends in user preferences or dietary restrictions.

The distribution in Figure 1 shows that most of the

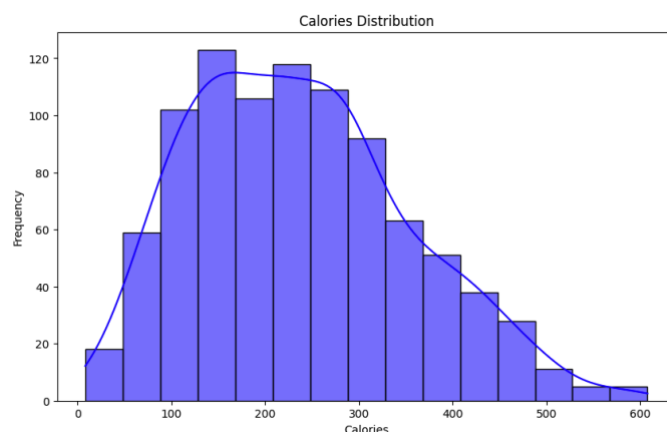


Figure 1: Calories Distribution

recipes fall in the [100, 350] calories bracket. There is no extreme skewness in data but there is just a little hint of right skewness with tail towards recipes of 600 calories.

Next, I analyzed the recipe key distribution, which involved examining the frequency of different dietary categories like "Gluten-Free," "Low-Carb," and "Vegetarian." This step allowed me to get insights into the diversity of the dataset.

In Figure 2, I have used bar plot instead of histograms

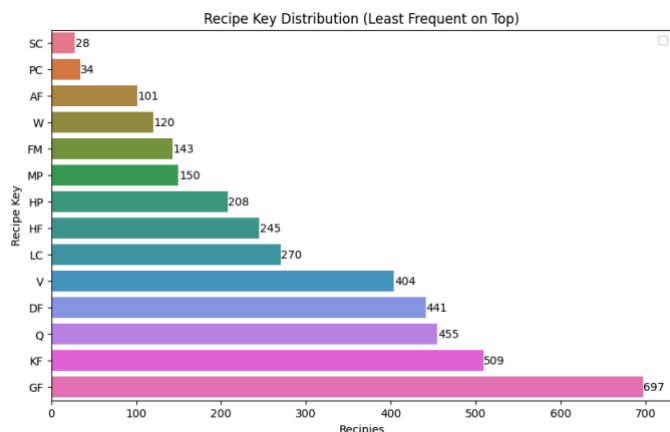


Figure 2: Recipe Keys Distribution

because we don't need bins here. We have discrete values for recipe keys, and I wanted to see how many recipes belong to each key in a sorted manner. As can be seen in Figure 2, Slow Cooker recipes had the least count while Gluten Free had the maximum count.

I also performed an analysis of personal points distribution to understand the variation in points assigned to each recipe.

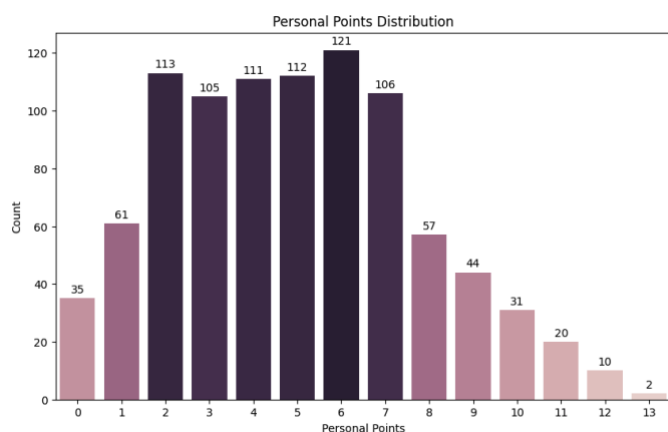


Figure 3: Personal Point Distribution

In Figure 3, It can be seen that most of the recipes have points between 2 and 7. I can't say much about this feature because of a lack of feature understanding.

Together, these analyses provided a comprehensive overview of the nutritional content and classification of the recipes.

7. User Interaction Component

A key component of this project was developing a user interaction interface that would allow users to filter the recipe data based on their preferences. The user interaction component is designed to take input for a calorie range and a personal points range, allowing users to tailor their search to recipes that fit within their dietary goals. Once the ranges are input, the program outputs the first 10 foods sorted by calorie content. For each recipe, the program displays the image of the food and a summary, providing users with a quick overview of the recipe's nutritional content and its overall appeal. This component enhances the usability of the dataset, allowing users to engage directly with the data and make informed decisions based on their preferences. It also highlights the practical applications of the scraped data and demonstrates how it can be used to offer personalized recommendations.

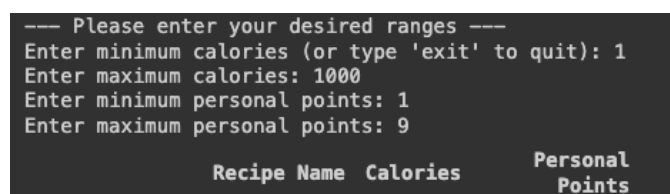


Figure 4: Snapshot of User Interaction

8. Conclusion

In conclusion, this mini-project allowed me to successfully scrape recipe data from www.skinnytaste.com, clean and preprocess the data, and perform exploratory data analysis (EDA). The process of scraping, cleaning, and analyzing the data provided invaluable hands-on experience in web scraping, data preprocessing, and EDA techniques. The user interaction component added a layer of practicality to the project, demonstrating how the dataset could be used for personalized recommendations based on user preferences.

However, there are some limitations to this mini-project. Since I targeted specific elements on the website using their class attributes, any changes made by the site developers to these classes would require adjustments in the codebase. This reliance on the structure of the website introduces a level of fragility, as scraping scripts may break if the HTML structure changes.

Moreover, this project deepened my understanding of the challenges involved in web scraping, particularly in distinguishing relevant content from non-relevant content and dealing with unstructured data. Overall, the project served as a comprehensive exercise in data collection and analysis, with the potential for further machine learning applications.